Generating Synthetic Supernovae Using Generative Adversarial Networks

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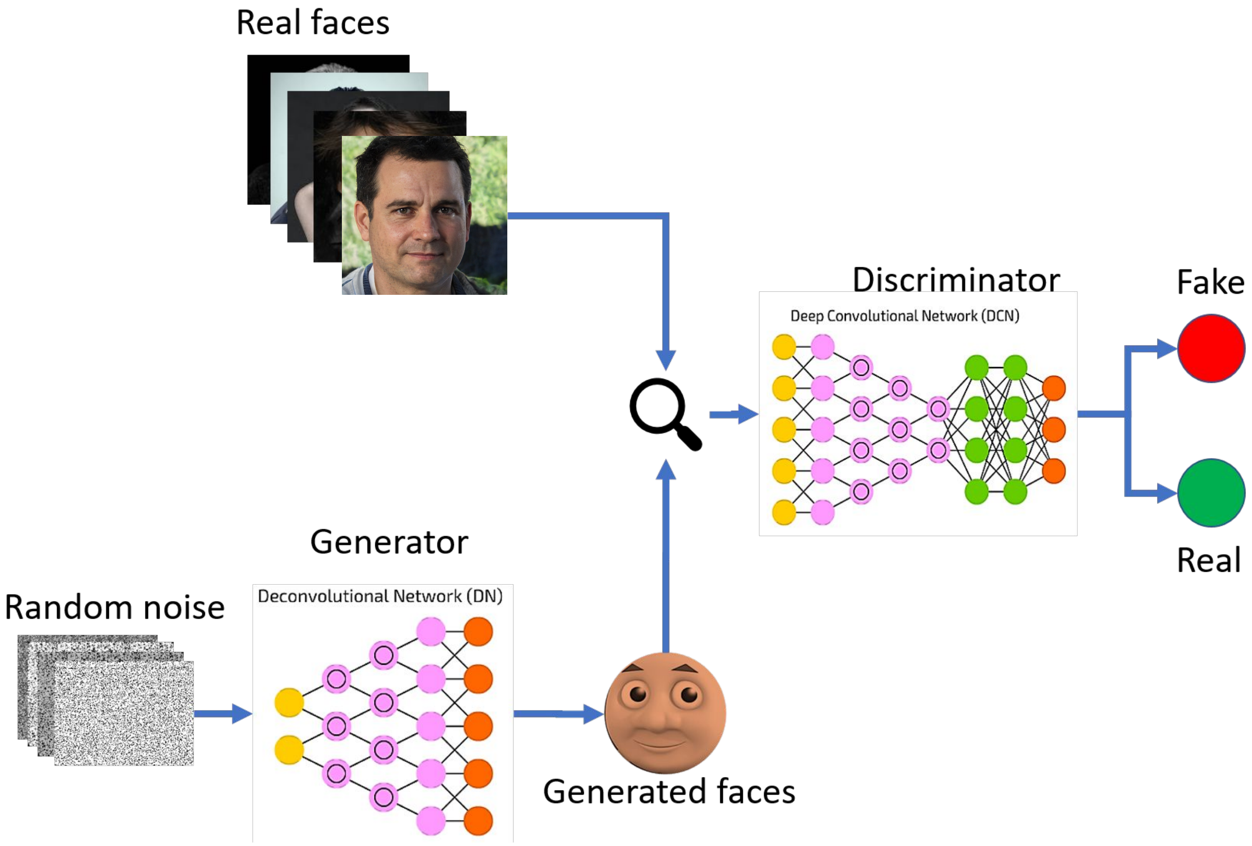
Generative Adversarial Networks (GANs)

Credit: https://thispersondoesnotexist.com/

Credit: https://thiscatdoesnotexist.com/

Generative Adversarial Networks (GANs)



Motivation

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- Training photometric classifiers requires large samples of labelled supernova light curves
- Physical simulations are limited by our understanding of the processes involved
- GANs provide a data-driven approach for generating supernova light curves, able to learn from a data-set without any assumptions about the physics involved
- Augmenting data-sets using GANs has been shown to improve classification performance within astronomy and beyond (e.g. Motamed+21, Garcia-Jara+22)
- Once a GAN is trained, thousands of light curves can be generated in seconds

Our Approach

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- Train a GAN using recurrent neural networks (RNNs) for both generator and discriminator
- RNNs perfectly suited for time series data, allows for generation of variable length time series
- We train a GAN based on samples of light curves from Maria Vincenzi’s DES simulations
- Our model generates observer-frame DES-like *griz* light curves

Light Curve Examples (SNe II)

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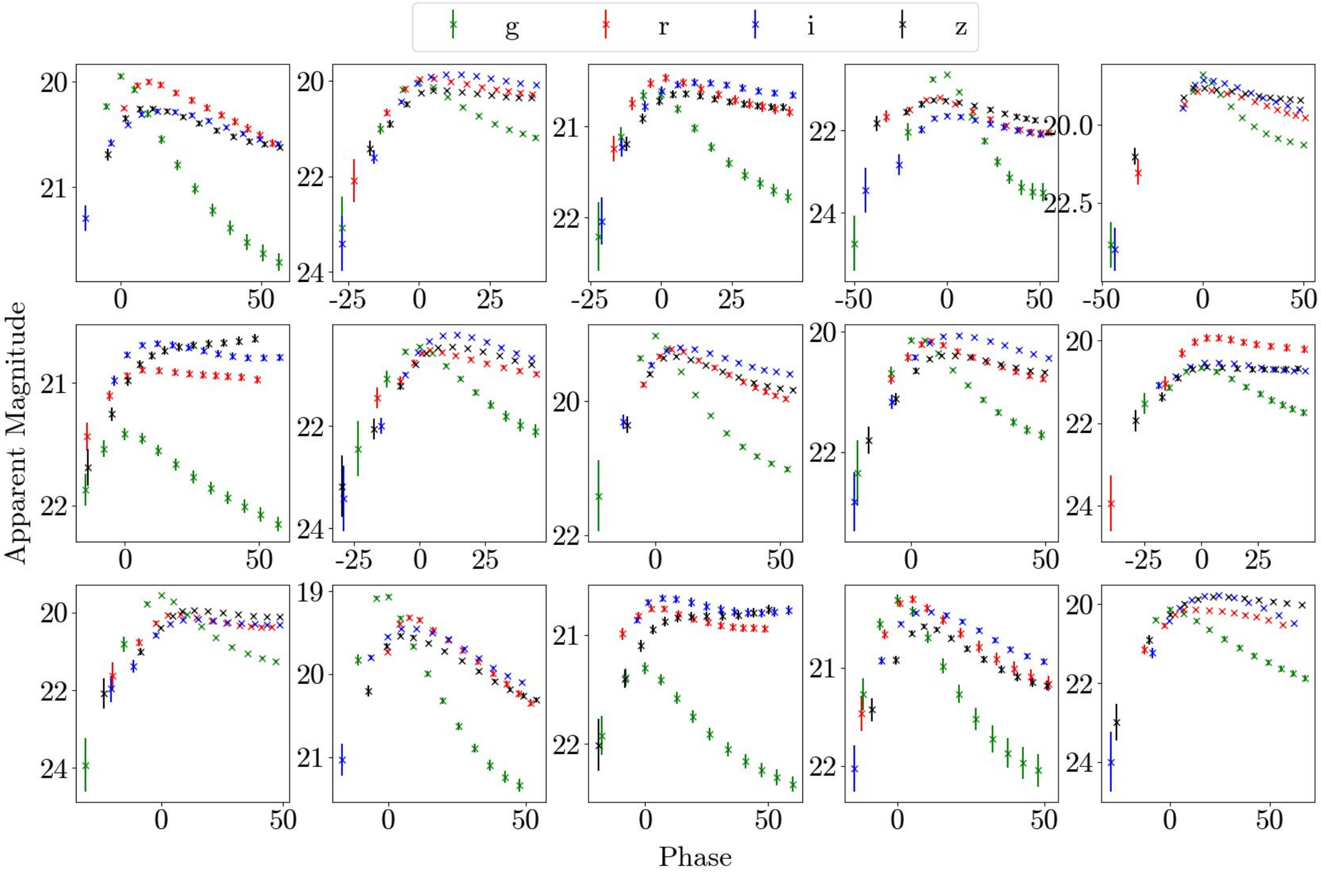

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Peak Apparent Magnitude

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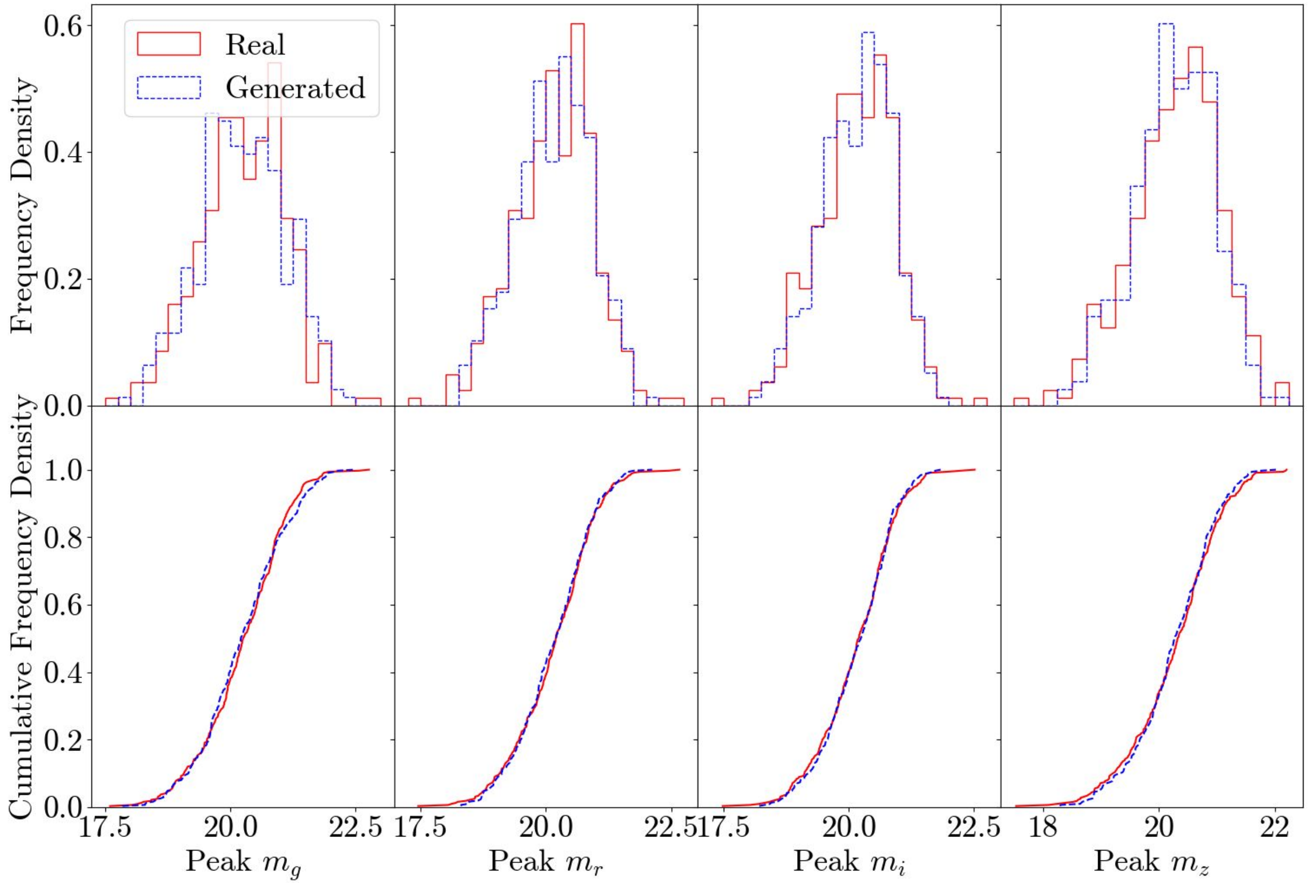

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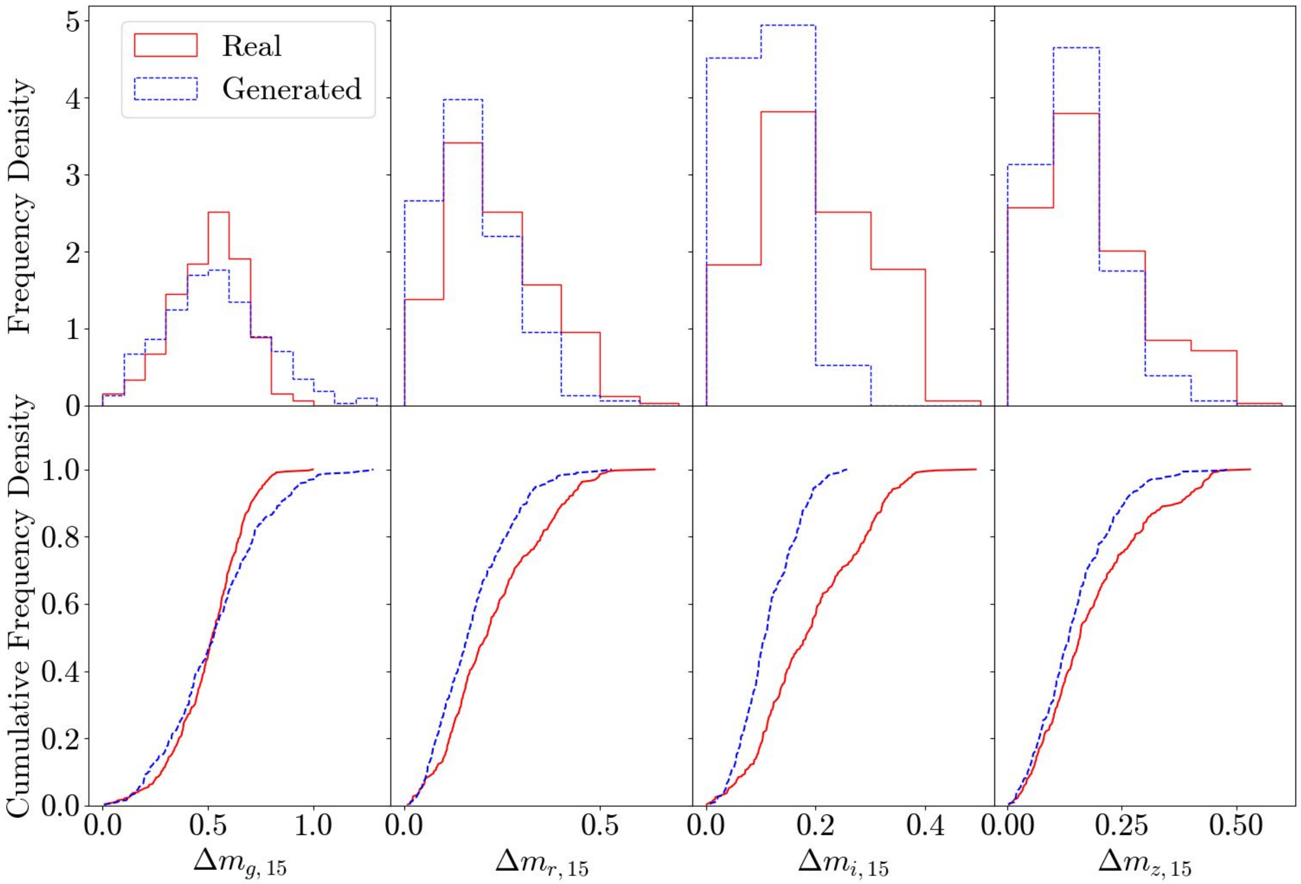
15 Day Declines



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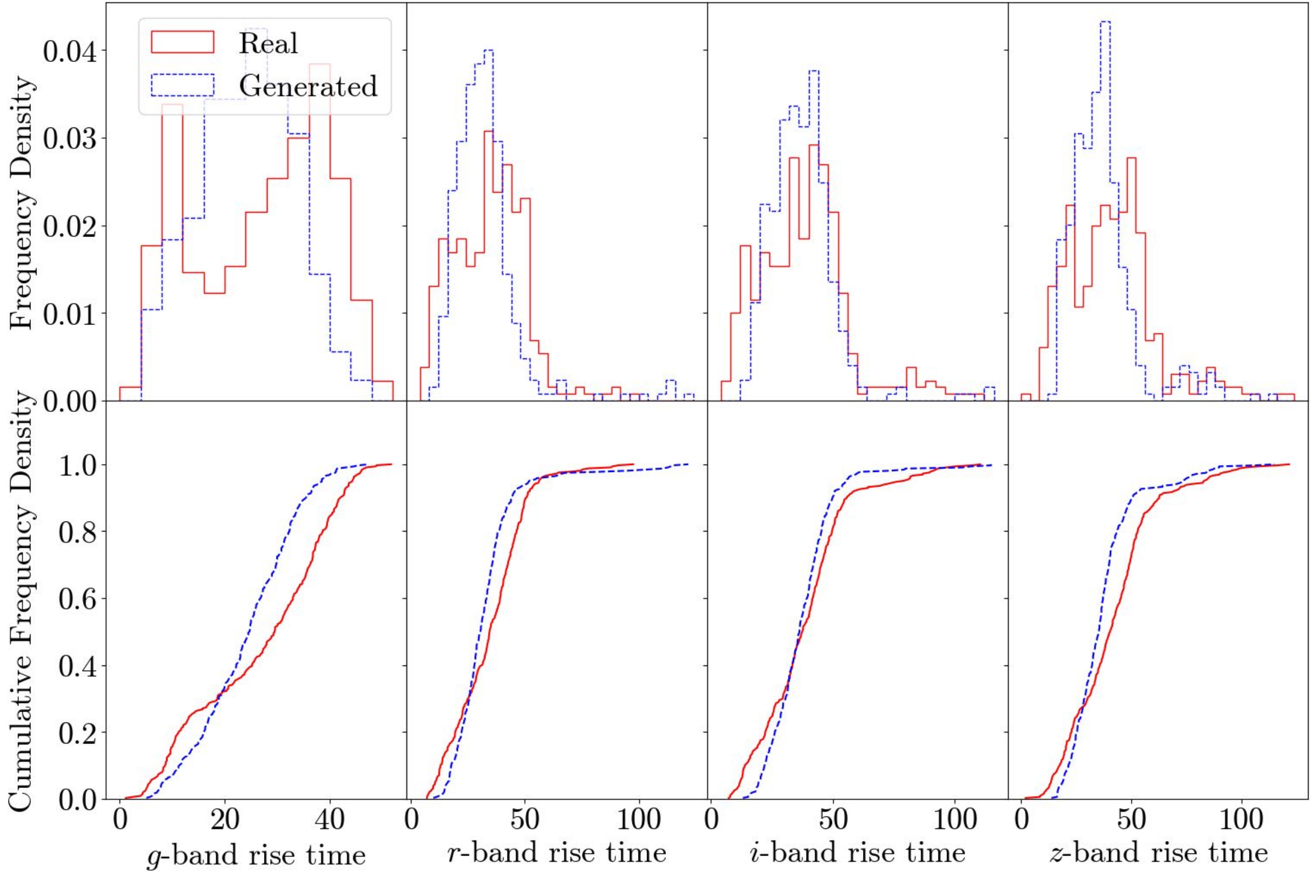
Rise times



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Mean Light Curves

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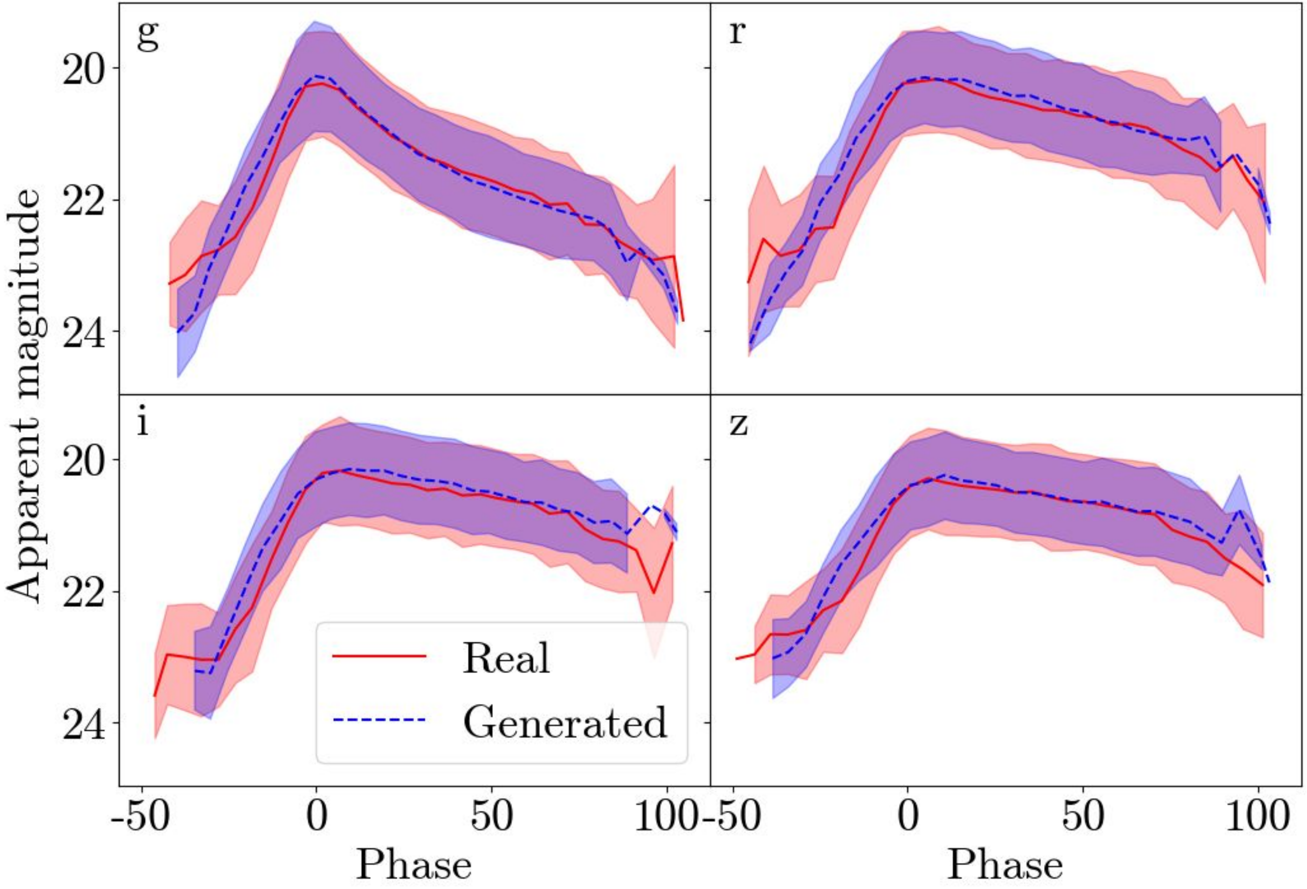

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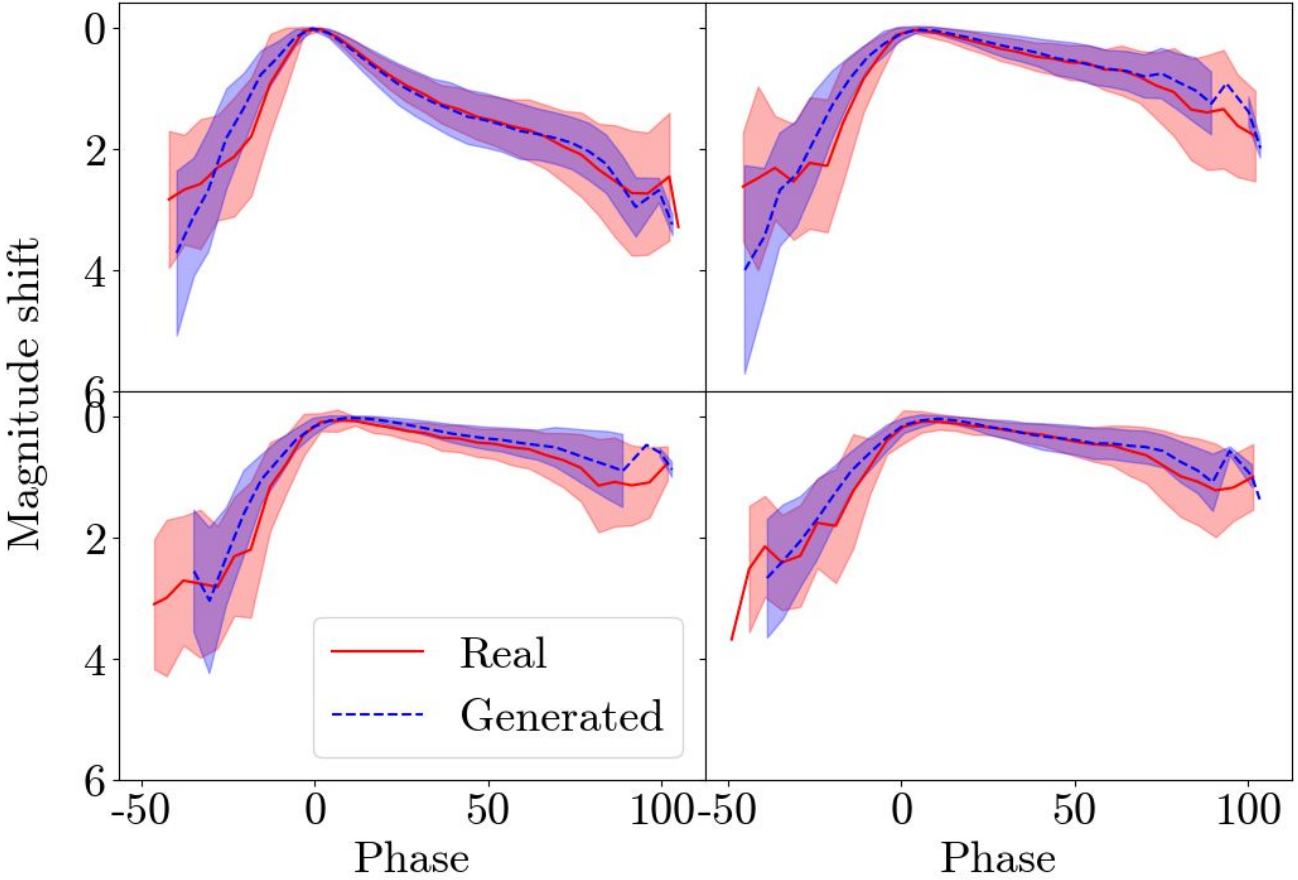
Mean Light Curves (shift from peak)



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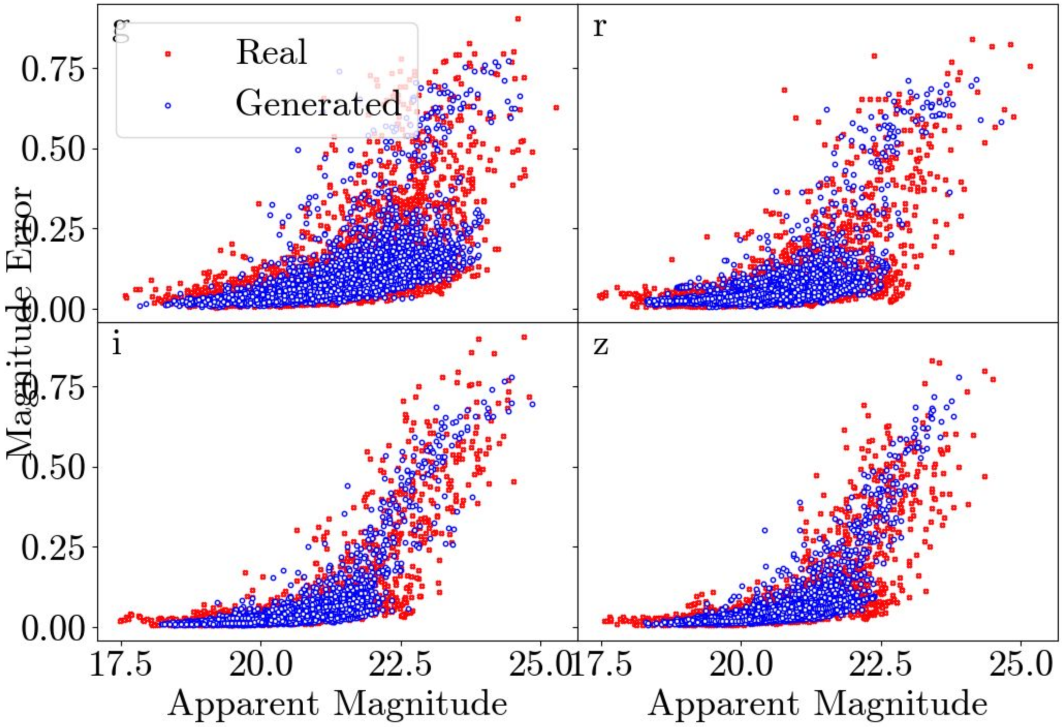
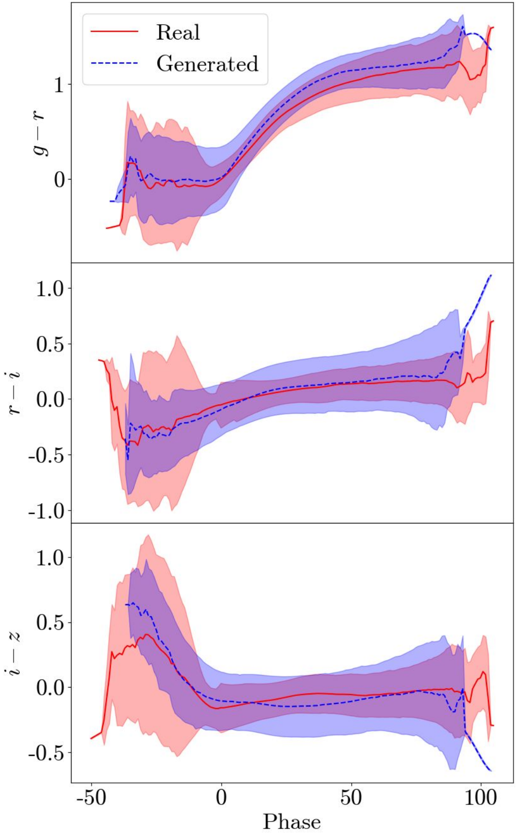
Errors and Colour Curves





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Summary

- GANs can be used to create large samples based on a small training set
- Work is ongoing but this approach is able to generate physically realistic light curves
- This technique can help to improve photometric classification, and can be used for any SN type and survey
- Code and paper to be published later this year